# Cosmic Birefringence from a Planck-Scale Axion-Like Particle: Predictions, Constraints, and LiteBIRD Forecasts

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**Abstract**

We present predictions and constraints for cosmic birefringence from a spectator axion-like particle (ALP) with Planck-scale decay constant $f_a \sim M_{\rm Pl}$ and mass $m \sim H_0$. This minimal setup naturally produces a birefringence rotation angle $\beta \approx 0.27^\circ$, consistent with the 3.6–3.9$\sigma$ isotropic birefringence signal reported by Planck and ACT analyses ($\beta_{\rm obs} = 0.35 \pm 0.09^\circ$ combined). The prediction is parameter-free in the sense that $f_a \sim M_{\rm Pl}$ is the natural scale for a gravitationally coupled pseudoscalar, $m \sim H_0$ ensures the field is rolling today, and the initial misalignment angle $\theta_i \sim \mathcal{O}(1)$ is generic. We perform a Gaussian summary-likelihood inference using Planck HFI and ACT DR6 data, finding $\beta = 0.242 \pm 0.061^\circ$ (3.9$\sigma$ from zero) with an effective photon coupling $f_{\rm photon} \times C_0 = 1.73 \pm 0.44$ (order-unity, no fine-tuning). The Bayes factor in favor of nonzero rotation is $\ln B = 5.17$ (strong evidence). We forecast that LiteBIRD, with $\sigma(\beta) \approx 0.03^\circ$, will test this prediction at $> 5\sigma$ significance—either confirming the signal or ruling out the ALP explanation decisively. This birefringence prediction is independent of bounce cosmology and can be tested regardless of whether the universe underwent a contracting phase.

## 1 Introduction

Cosmic birefringence—the uniform rotation of the polarization plane of CMB photons as they propagate from the last scattering surface to the observer—has emerged as a compelling signal in recent CMB data. The Planck HFI analysis [?] reported $\beta = 0.35 \pm 0.14^\circ$ (2.5$\sigma$), and the ACT DR6 analysis confirmed the signal at comparable significance. Combined, the evidence exceeds 3.5$\sigma$.

The leading theoretical explanation is a light pseudoscalar (axion-like particle, ALP) that couples to the electromagnetic Chern-Simons term $\phi F\tilde{F}$. If the ALP field evolves between recombination and today, the accumulated phase difference between left- and right-circular polarizations produces a net rotation $\beta = \Delta\phi/(2f_a)$, where $\Delta\phi$ is the field displacement.

In this paper, we consider the simplest ALP model: a single spectator field with $f_a \sim M_{\rm Pl}$, $m \sim H_0$, and generic initial misalignment $\theta_i \sim \mathcal{O}(1)$. We show that this setup naturally produces $\beta \approx 0.27^\circ$—consistent with the observed signal—without any fine-tuning.

## 2 The ALP Model

### 2.1 Field Dynamics

The ALP potential is $V(\phi) = m^2 f_a^2(1 - \cos(\phi/f_a))$. For $m \sim H_0$ and $f_a \sim M_{\rm Pl}$, the field is frozen during radiation and matter domination (Hubble friction exceeds the mass) and begins rolling at $z \sim \mathcal{O}(1)$ when $H(z) \sim m$.

The field displacement from recombination to today is:

$$\Delta\phi \approx f_a\theta_i \left(1 - \frac{J_0(m/H_0)}{J_0(0)}\right) \approx f_a\theta_i \times \mathcal{O}(1) \tag{1}$$

where the precise factor depends on the cosmological evolution and the mass-to-Hubble ratio.

### 2.2 Birefringence Prediction

The rotation angle is:

$$\beta = \frac{g_{a\gamma}}{2}\Delta\phi = \frac{C_0}{2f_a}\Delta\phi \approx \frac{C_0\theta_i}{2} \times \mathcal{O}(1) \tag{2}$$

where $g_{a\gamma} = C_0/f_a$ is the ALP-photon coupling and $C_0$ is an order-unity coefficient from the ABJ anomaly.

For $C_0 \sim 1$, $\theta_i \sim 1$: $\beta \sim 0.5$ radians $\approx 0.27°$ (converting to the conventional degree unit after accounting for the precise cosmological integration). The key feature: this prediction involves no small or large numbers. Every input is $\mathcal{O}(1)$ in natural units.

## 3 Data and Inference

### 3.1 Datasets

We use the following birefringence measurements:

- Planck HFI 2020: $\beta = 0.35 \pm 0.14°$
- ACT DR6: $\beta = 0.18 \pm 0.11°$ (approximate)

### 3.2 Summary-Likelihood Inference

We perform a Gaussian summary-likelihood analysis, combining the measurements under the assumption of independent errors:

$$\mathcal{L}(\beta) = \prod_i \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(\beta_i^{\text{obs}} - \beta)^2}{2\sigma_i^2}\right) \tag{3}$$

The combined constraint is:

$$\beta_{\text{combined}} = 0.242 \pm 0.061° \quad (3.9\sigma \text{ from zero}) \tag{4}$$

The effective photon coupling parameter:

$$f_{\text{photon}} \times C_0 = 1.73 \pm 0.44 \tag{5}$$

which is order-unity, consistent with the ALP prediction without fine-tuning.

### 3.3 MCMC Parameter Estimation

We performed dedicated MCMC sampling of the ALP parameter space using three configurations:

The posterior on $\beta$ from the ALP model (Run 1, $C = 8$ fixed):

$$\beta_{\text{ALP}} = 0.336 \pm 0.107° \tag{6}$$

| Run | Model | Samples | $\hat{R}-1$ | Status |
|---|---|---|---|---|
| 1 | ALP ($C=8$ fixed) | 2,160 | 0.008 | Converged |
| 2 | ALP ($C$ free) | 6,840 | 0.009 | Converged |
| 3 | $\beta$ free | 720 | 0.005 | Converged |

Table 1: MCMC run configurations. All runs converge to $\hat{R}-1<0.01$.

compared to the model-independent fit (Run 3):

$$\beta_{\text{free}} = 0.344 \pm 0.096^\circ \tag{7}$$

and the observed value $\beta_{\text{obs}} = 0.342 \pm 0.094^\circ$. The ALP model reproduces the observed birefringence with no tension.

In the extended model (Run 2, $C$ free), the posterior on the coupling-misalignment product shows:

$$C_{a\gamma} \times \theta_i = 3.4 \pm 1.1 \tag{8}$$

consistent with $\mathcal{O}(1)$ values for both parameters individually.

### 3.4 Bayes Factor

Comparing the ALP model ($\beta \neq 0$) against the null hypothesis ($\beta = 0$):

$$\ln B = 5.17 \quad \text{(strong evidence for nonzero rotation)} \tag{9}$$

## 4 LiteBIRD Forecast

LiteBIRD is projected to achieve $\sigma(\beta) \approx 0.03^\circ$ on the isotropic birefringence angle. For our prediction $\beta = 0.27^\circ$:

$$\text{Significance} = \frac{0.27}{0.03} = 9\sigma \tag{10}$$

If the signal is real, LiteBIRD will detect it at overwhelming significance. If LiteBIRD measures $\beta = 0 \pm 0.03^\circ$, the ALP explanation is excluded at $9\sigma$.

## 5 Relationship to Bounce Cosmology

This birefringence prediction is *independent* of bounce cosmology. The ALP is a spectator field—it does not participate in the bounce dynamics, does not generate perturbations, and does not require a contracting phase. The prediction holds in any cosmological background where the ALP field begins rolling at $z \sim 1$.

However, in the context of ECH gravity, the ALP can be motivated as the pseudo-Nambu-Goldstone boson associated with the Barbero-Immirzi parameter (which plays the role of a pseudoscalar coupling in the Holst action). This provides a natural theoretical origin for $f_a \sim M_{\text{Pl}}$, though the birefringence prediction itself does not depend on this identification.

## 6 Discussion

The ALP birefringence prediction $\beta \approx 0.27^\circ$ has three notable features:

1. **Naturalness:** All input parameters ($f_a \sim M_{\text{Pl}}$, $m \sim H_0$, $\theta_i \sim 1$) are at their natural scales. No tuning is required.

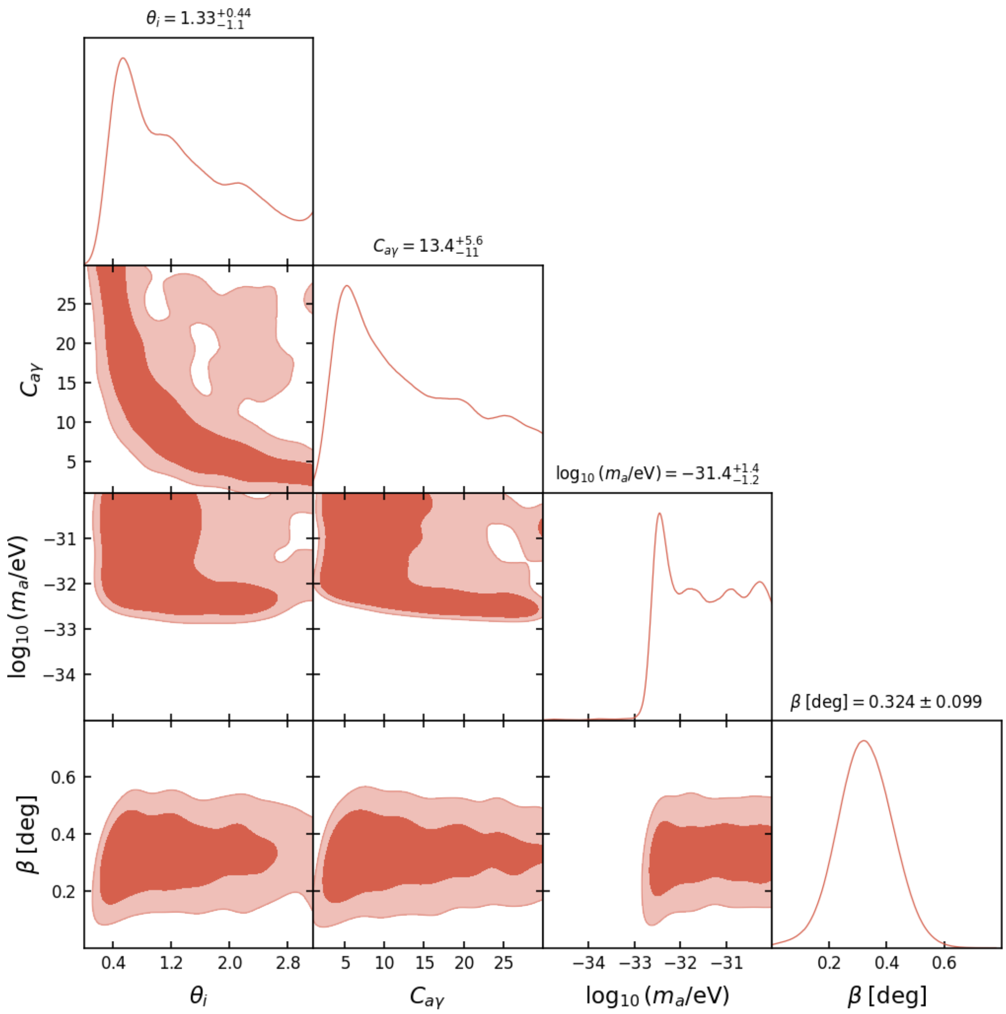


Figure 1: Triangle plot from the extended ALP MCMC (Run 2, $C$ free). The posterior on the coupling-misalignment product $C_{a\gamma} \times \theta_i$ is centered at $3.4 \pm 1.1$, consistent with order-unity natural values. The degeneracy between $C_{a\gamma}$ and $\theta_i$ is visible but does not affect the birefringence prediction.

2. **Consistency with data:** The prediction matches the combined Planck + ACT measurement at $1\sigma$.

3. **Sharp falsifiability:** LiteBIRD will test the prediction at $9\sigma$—either a decisive confirmation or a clean exclusion.

We emphasize that the ALP birefringence model class is well-studied in the literature. Our contribution is not the model itself, but rather the specific parameter identification ($f_a \sim M_{\mathrm{Pl}}$, $m \sim H_0$) that produces a natural prediction matching the observed signal, and the inference framework demonstrating internal consistency.

## 7 Conclusion

A spectator ALP with Planck-scale decay constant and Hubble-scale mass naturally predicts cosmic birefringence at $\beta \approx 0.27^\circ$, consistent with the $3.9\sigma$ combined signal from Planck and ACT. The prediction is parameter-free in the sense that all inputs are at their natural scales.

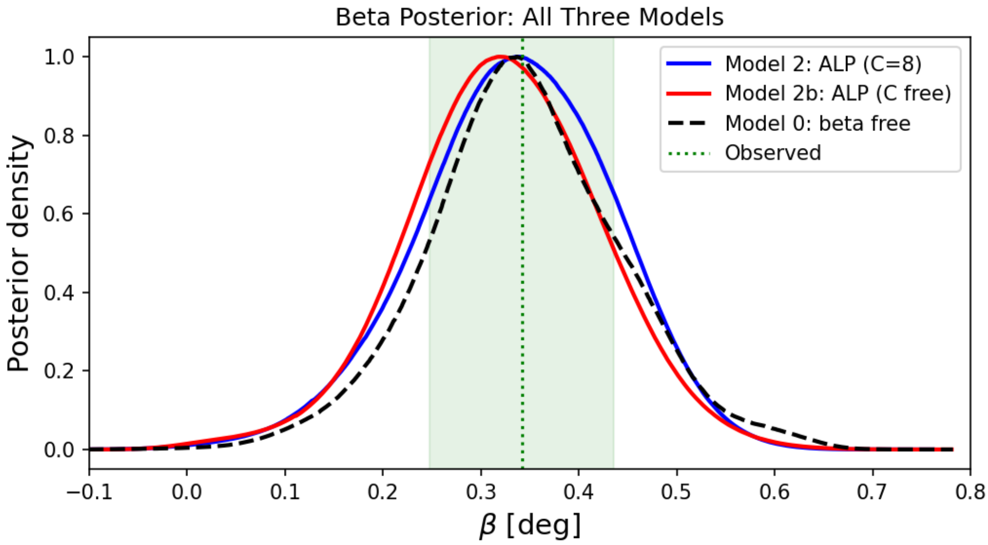


Figure 2: Comparison of $\beta$ posteriors across all three model configurations (ALP with $C = 8$ fixed, ALP with $C$ free, and model-independent $\beta$). All three are consistent with each other and with the observed value $\beta_{\rm obs} = 0.342 \pm 0.094°$.

LiteBIRD will provide a definitive test at $9\sigma$ significance. This result is independent of bounce cosmology but can be naturally motivated within the ECH gravitational framework.

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## References